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Development of an Integrated Solar Heat Pump Concept using Ice Slurry as a Latent Storage Material

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Abstract

This paper presents the design analysis of a solar heat pump using cool thermal storage. The integrated concept uses cool storage (sensible or ice-based) as a short-term (i.e. days, weeks) storage for solar energy in winter and cooling energy in the summer. Annual TRNSYS simulations are performed in four Canadian regions to identify the potential of ice storage and unglazed solar collectors. Results show that combining these two technologies can reduce mechanical system energy use by 17% - 28% compared to a cold-climate air-water heat pump. Unglazed collectors demonstrated potential in areas with milder winters.

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1. Introduction

The Canadian residential sector accounts for 16% of national secondary energy use, with over 80% of this total directed towards space heating, cooling, and domestic hot water (DHW)¹. Heat pumps can significantly reduce residential energy consumption by using renewable energy sources to efficiently meet building thermal demands. Currently, the majority of residential heat pumps in North America use air as the main source of thermal energy². However, these units can lose a significant portion of heating capacity below 0°C, posing an issue in cold Canadian climates³. As such, there is a strong need to leverage other renewable energy sources to consistently meet building thermal demands during even the coldest winter days.

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Nomenclature

α	Solar absorptivity (-)
Δ	Temperature difference ($^{\circ}\text{C}$)
ε	Solar emissivity (-)
ACH ₅₀	Air changes per hour @ 50 Pa
Col	Solar collector
COP	Coefficient of performance (-)
DB	Dry bulb
DHW	Domestic hot water
Fluid	Fluid phase
HRV	Heat recovery ventilator
I	Initial
IT	Ice tank
M	Mass (kg)
N	Number of panels
Max	Maximum capacity of tank
SAHP	Solar assisted heat pump
T	Temperature ($^{\circ}\text{C}$)
RF	Radiant floor
WB	Wet bulb
WT	Warm tank

Solar assisted heat pumps (SAHP) have shown promise in using low grade thermal energy to supplement and improve heat pump performance in cold climates⁴. Despite this strong energy savings potential, these systems face several challenges, including a lack of adequate thermal storage capacity and reduced solar gains in the winter months. In Canada, a growing demand for space cooling places an additional requirement on system functionality. Addressing these issues in a simple and effective manner is critical to the market acceptance of these systems.

Combining solar heat pumps with ice-based latent storage can significantly improve energy storage densities, while the use of a cold temperature working fluid in the solar collectors can increase thermal gains and extend collector utilization periods. Although several system configurations have been proposed for locations both in Europe and North America^{5,6,7,8} there has been little analysis to quantify the impact of ice storage and identify the most effective configurations for different climates.

This paper presents a Canada-wide analysis on the role of ice storage and unglazed solar collectors for an innovative solar heat pump system. A simple cooling mode is also proposed in which a cool storage tank, charged during summer DHW operations, meets space cooling requirements for the building. Separate high performance housing models are developed for four Canadian regions and used to establish baseline energy consumption. Three distinct systems are then examined to identify the impact of ice storage and solar collector type on annual energy use. Finally, performance is presented on a system and component level to better assess each proposed design.

2. Development of High Performance Housing Models

Housing models are developed for four Canadian regions to identify the impact of climate on system energy performance. Table 1 summarizes key climate parameters for each region^{9,10,11}. Heating and cooling degree days use a balance temperature of 18 $^{\circ}\text{C}$, while solar potential is shown for a surface tilted to the latitude of the location.

In Halifax, the Atlantic Ocean moderates temperatures, resulting in cloudier conditions with milder winters (in comparison to central Canada) and cool summers. Montreal represents a continental climate typical of central Canada, with cold winters and warm summers yielding significant space heating and cooling requirements. In Calgary, although winter temperatures can be quite cold, solar potential is high, making it an interesting candidate for solar-based systems. Finally a cool-marine climate prevails in Vancouver, with lower solar potential but a relatively high average annual air temperature near 10 $^{\circ}\text{C}$.

Table 1. Climate summary.

	Halifax	Montreal	Calgary	Vancouver
Latitude	N 44°	N 45°	N 51°	N 49°
Heating Degree Days	4000	4200	5000	2950
Cooling Degree Days	207	426	217	153
Heat T _{Design} (°C,DB)	-16	-23	-30	-7
Cool T _{Design} (°C,DB/WB)	26/20	30/23	28/17	28/19
Annual Solar (kWh/m ²)	1074	1185	1292	1009

Developed housing models were based on the Canadian Centre for Housing Technology (CCHT) test home¹². The CCHT home is typical of single family housing in Canada, and consists of two above ground floors and a finished basement with a total heated floor area of 284 m². The building envelope and mechanical systems were modified in each region to represent a Net Zero Ready (NZR) home, which can be defined as a home having the building envelope and infrastructure such that it becomes cost effective to integrate onsite renewable generation¹³. For this study, a NZR home was defined as achieving an ERS-86 rating on the EnerGuide Rating Scale¹⁴. Table 2 summarizes building envelope parameters by region.

Table 2. Key housing envelope parameters.

	Halifax	Montreal	Calgary	Vancouver
Roof RSI	8.93 m ² °C/W	8.93 m ² °C/W	8.93 m ² °C/W	8.93 m ² °C/W
Wall RSI	4.83 m ² °C/W	5.46 m ² °C/W	5.68 m ² °C/W	4.48 m ² °C/W
Basement Wall RSI	4.95 m ² °C/W	4.95 m ² °C/W	4.95 m ² °C/W	4.95 m ² °C/W
Basement Slab RSI	2.58 m ² °C/W	2.58 m ² °C/W	2.58 m ² °C/W	1.86 m ² °C/W
Window U-Value	1.01 W/m ² °C	1.01 W/m ² °C	1.01 W/m ² °C	1.01 W/m ² °C
Infiltration	1.00 ACH ₅₀	0.75 ACH ₅₀	0.75 ACH ₅₀	1.00 ACH ₅₀

To assess solar system performance it was also important to define two base mechanical systems representing conventional and more efficient heating technologies. Table 3 summarizes key details for each system. In all cases, DHW draws are set to 233 L/day. A split system was used for cooling in Base Case 2, as the air-water heat pump operates only in heating mode to meet heating/DHW loads year round. Tank volumes for Base Case 2 are larger as this tank serves both DHW and radiant flooring loops. Heat pump performance was based on manufacturer data¹⁵.

Table 3. Base case mechanical systems.

	Base Case #1	Base Case #2
Heating System	Electric Baseboard	Air-Water Heat Pump (COP 3.70 [†]), Radiant Flooring
Cooling System	Split System	Split System
	Rated COP 3.45 [‡]	Rated COP 3.45 [‡]
Ventilation	HRV, 0.84 effectiveness	HRV, 0.84 effectiveness
DHW	Electric Conventional Tank	Air-Water Heat Pump + Tankless Electric (if needed)
	Tank Volume 0.23 m ³	Tank Volume 0.50 m ³ (Serves RF and DHW loops)

[†] Rated at 8.3°C outdoor dry bulb, 30°C inlet water temperature

[‡] At indoor temperature of 27°C DB/19°C WB, 35°C outdoor temperature

3. Proposed Solar Heat Pump Concept

Two technologies are central to the solar-based systems presented in this study:

1. **Cool Storage.** Cool storage (ice-based latent, or sensible only) provides cold temperature fluid to circulate to the collectors. This minimizes thermal losses to the outside air, allowing for more efficient, sustained collector operations. When ice storage is used, the latent heat available during the ice/water phase change also allows for increased energy storage densities.
2. **Heat Pump.** Heat pumps provide an efficient way to transfer low grade thermal energy obtained by the solar collectors for use in heating and DHW systems. Combining a heat pump with ice storage also improves heat pump performance by providing a stable minimum source temperature of 0°C. This is especially true in comparison to air-source systems, as outdoor temperatures in cold climates can fall well below 0°C during the winter months.

A common diagram for all proposed system variations is provided in Fig. 1. The proposed concept is similar to Tamasauskas *et al.*⁸, except that only one solar loop is used in order to simplify operations and minimize required equipment. Solar collectors are the main source of thermal energy to the system, while a heat pump is used to deliver collected energy to a warm tank serving radiant flooring and DHW loops. Additional heating requirements can be met using auxiliary fluid heaters in the radiant flooring loops and DHW loop.

A simple cooling mode is proposed for the warmer months of the year (May 1 to Sept 30). By limiting solar gains into the cool tank during these months, cooling capacity can be built in the cool tank during normal heat pump DHW operations. Cool fluid from the cool tank can then be used as a source for a central cooling coil. Should temperatures in the cool tank become too warm, the heat pump can also reject thermal energy to the outdoors via an air-cooled condenser. This mode adds an important degree of system functionality, especially in regions with significant cooling demands.

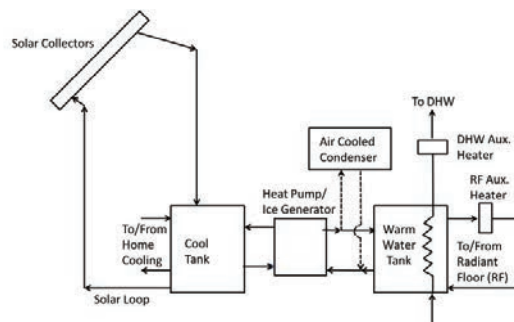


Fig. 1. Layout for proposed solar systems.

3.1. System Variations

Table 4 summarizes each system variation examined in this study. Each case is designed to identify the impact of ice storage and solar collector type on overall performance. Ice storage is based on the use of ice slurry (a mixture of small diameter ice particles and water) as described in Tamasauskas *et al.*⁸.

Unglazed collectors are examined due to their simplicity and relatively low cost. These collectors also have a higher efficiency than glazed collectors when temperature differences between the collector fluid and outdoor air are small, which has important implications for ice-based storage systems.

Table 4. Proposed system variations.

	Collector Type	Storage Type
SAHP #1	Glazed, flat plate	Cool, sensible-only
SAHP #2	Glazed, flat plate	Ice-based
SAHP #3	Unglazed, flat plate	Ice based

3.2. System Operations and Control

Solar Loop Control

Solar loop operations are based on the following variables:

1. *Useful solar gains*. An initial predicted temperature rise ($\Delta T_{Col,I}$) of 5°C is required to start solar loop operations. This ensures that the collector loop operates only when sufficient solar gains are available, mitigating the risk of freezing during loop operations. Once started, loop operations continue as long as the fluid temperature rise is greater than 1°C. The loop also contains a drain-back feature to mitigate freezing potential when there is no fluid circulation.
2. *Cool tank fluid temperature*. Fluid temperatures in the cool storage ($T_{Fluid,IT}$) are limited to a maximum of 13°C, which is the maximum inlet evaporator temperature for the heat pump based on the compressor datasheet⁸.
3. *Available stored energy* (Cooling Season only). During the cooling season, solar loop operations are controlled to build up suitable cooling capacity for the building. For SAHP #1, the solar loop operates only when fluid temperatures in the cool tank fall below 5°C. For SAHP #2 and SAHP #3, the solar loop operates only when the ice mass is greater than 90% of the ice storage capacity ($M_{Ice,Max}$), with the maximum ice capacity set at 60% of the tank fluid mass based on experimental results from a test bench⁸

Heat Pump Control

Heat pump operations are based on:

1. *Available stored energy*. For SAHP #1, the heat pump operates only when cool tank fluid temperatures are above 4.4°C (the minimum inlet temperature for a typical water-source unit operating without glycol). For SAHP #2 & SAHP #3, the heat pump operates when the ice mass is below the maximum ice tank capacity.
2. *Warm tank temperatures*. The heat pump maintains a 45°C temperature at the top of the warm tank ($T_{Fluid,WT}$).
3. *Cool tank fluid temperature* (Cooling Season Only). In the cooling season, the heat pump also operates when cool tank fluid temperatures exceed 7°C. This mode is in addition to heating operations based on Heat Pump Condition 2. Thermal energy is either rejected to the warm tank (if $T_{Fluid,WT} < 45^\circ\text{C}$), or the ambient via the air-cooled condenser.

A summary of conditions necessary to begin solar or heat pump operations is provided in Table 5.

Table 5. Summary of solar and heat pump loop starting conditions.

		SAHP #1	SAHP #2, SAHP #3
Solar Loop	Heat Season	$\Delta T_{Col,I} > 5^\circ\text{C}$, $T_{Fluid,IT} < 13^\circ\text{C}$	$\Delta T_{Col,I} > 5^\circ\text{C}$, $T_{Fluid,IT} < 13^\circ\text{C}$
	Cool Season	$\Delta T_{Col,I} > 5^\circ\text{C}$, $T_{Fluid,IT} < 5^\circ\text{C}$	$\Delta T_{Col,I} > 5^\circ\text{C}$, $T_{Fluid,IT} < 13^\circ\text{C}$, $M_{Ice} > 0.9M_{Ice,Max}$
Heat Pump	Heat Season	$T_{Fluid,IT} > 4.4^\circ\text{C}$, $T_{Fluid,WT} < 45^\circ\text{C}$	$M_{Ice} < M_{Ice,Max}$, $T_{Fluid,WT} < 45^\circ\text{C}$
	Cool Season	$T_{Fluid,IT} > 4.4^\circ\text{C}$, $T_{Fluid,WT} < 45^\circ\text{C}$ or $T_{Fluid,IT} > 7^\circ\text{C}$	$M_{Ice} < M_{Ice,Max}$, $T_{Fluid,WT} < 45^\circ\text{C}$ or $T_{Fluid,IT} > 7^\circ\text{C}$

Heating Loop

Space heating demands are met using radiant flooring loops on each level of the home. Space setpoints are 21°C for the two upper floors, and 18°C in the basement. An outside air temperature reset linearly varies the supply inlet fluid temperature from 40°C at the heating design temperature (Table 1) to 25°C when the outside temperature is 18°C.

Cooling Loop

The home employs a two level cooling strategy. First, natural ventilation (via operable windows) is used to cool when space temperatures exceed 23°C, provided that outdoor temperatures are lower than those inside the home. Mechanical cooling is subsequently activated when space temperatures exceed 24°C, and is supplied via a fan coil located in the stairwell between the first and second floors. This fan coil configuration has been demonstrated both via simulation and practice to provide sufficient cooling to the upper two floors of the home (the basement cooling demand is negligible) in typical Canadian residences such as the one in this study¹⁶.

4. Simulation Methodology

TRNSYS¹⁷ was used for all simulation work because of its flexibility and large number of solar-based components. All models were simulated at a time step of 3.75 minutes, with this small time step required to appropriately model system control decisions and promote solution convergence in TRNSYS. The choice of 3.75 minutes has also been specifically chosen because this time step is of the form $1/2^n$ hours (where n is a user-selected integer), which ensures that there are no issues with TRNSYS components writing data to external files¹⁸. Solar systems included an additional month of pre-simulation to reduce the impact of initial conditions.

Table 6 summarizes selected HVAC components used to simulate the SAHP cases. All homes were simulated using the detailed TRNSYS building model (Type 56). Heat pump data was derived from an experimental test bench⁸, while tank sizes balanced storage capacity with space constraints. Solar collector performance^{19,20} and pump selections²¹ were based on manufacturer data. Unglazed solar collector²⁰ used in the simulations contain a selective coating which maximizes absorptivity ($\alpha=0.94$) while minimizing emissivity from the collector surface ($\epsilon=0.18$).

Total array areas for the glazed and unglazed collectors were selected to be as similar as possible. This decision was made to provide a base of comparison between these two configurations, and to identify regions where significant potential cost savings could be obtained by replacing glazed collectors with simpler unglazed units.

Table 6. Summary of TRNSYS simulation components.

Component	TRNSYS Type	Notes
Heat Pump	Type 927	4 ton, $COP_{Rated}=4.28^\dagger$
Ice Generator	Equation Type	Derived from Guilpart and Fournaison ²²
Cool/Ice Tanks	Type 217	$Volume_{IT}=5.0 \text{ m}^3$, custom model ⁸
Warm Tank	Type 534	$Volume_{WT}=0.5 \text{ m}^3$, 4 nodes
Flat Plate Glazed Collector	Type 539	$Area=1.99 \text{ m}^2/\text{panel}$, $N=20$, $Slope=40^\circ$, south-facing
Unglazed Solar Collector	Type 563	$Area=2.01 \text{ m}^2/\text{panel}$, $N=20$, $Slope=40^\circ$, south-facing
Solar Pump	Type 654	Constant speed, 60W, 60 kg/h/m ² (glazed), 38 kg/h/m ² (unglazed)
Radiant Flooring	Type 56 Active Layer	Separate loops for all three levels
Heating Control	Type 23	Separate Proportional-Integral-Derivative (PID) control for each RF Loop

[†]Rated at 0°C evaporator temperature, 28°C condenser water temperature

5. Results

Annual simulations were performed to determine system energy performance by region. For all SAHP systems, cooling energy use refers to the operation of the air cooled condenser. Thermal energy rejected to the warm tank during the cooling season was counted for DHW purposes.

Table 7 presents annual system energy use for Halifax and Vancouver. For both cities, milder winter temperatures allow the air-water heat pump (Base #2) to achieve significant energy savings, with energy use reductions between 40% and 46%. Each solar heat pump system offers greater percentage energy use reductions in Halifax due to the larger solar potential, with savings between 19% and 28% in comparison to an air-water heat pump. While the impact of solar energy in Halifax is clear, adding ice storage results in a smaller 11% reduction in mechanical system energy use relative to the sensible only case. A closer examination of component results reveals that the ice capacity of the ice tank is used only briefly in December and January, somewhat limiting the benefit of ice storage in this location. In Vancouver, frequent cloudy conditions reduce collector gains, which subsequently results in a greater use of ice storage capabilities. While energy use for the sensible only case is quite similar to the air-water heat pump, both ice storage systems yield far greater savings of around 20%. In both locations, the milder winters allow unglazed collectors to perform effectively, providing a pathway towards low-cost system implementation. Unglazed collectors are particularly interesting in Vancouver because there is also the potential of obtaining convective gains in the winter when outdoor air temperatures are above collector fluid temperatures.

In both cities, using cool storage can greatly reduce cooling energy use. In fact, for Halifax and Vancouver, using the ice storage tank for cooling can completely eliminate the need for additional heat rejection via the air-cooled condenser, minimizing system complexity and cost.

Table 7. Annual system performance in Halifax and Vancouver.

	Halifax					Vancouver				
	Base #1	Base #2	SAHP #1	SAHP #2	SAHP #3	Base #1	Base #2	SAHP #1	SAHP #2	SAHP #3
Heat+DHW (kWh)	11151	5631	4518	3874	4054	10596	4597	4531	3587	3462
Cool (kWh)	146	146	43	0	0	69	69	30	0	0
Fans+Pumps (kWh)	972	1525	1363	1380	1387	968	1643	1453	1484	1493
Total Mech. (kWh)	12269	7302	5925	5254	5441	11632	6309	6015	5072	4954

Table 8 summarizes annual system energy performance for Montreal and Calgary. In both cities, the air-water heat pump (Base #2) substantially reduces system energy use, with savings between 43% and 46%. While the sensible only solar systems generally exhibit similar performance to the air-water heat pump, the true potential of the proposed layouts become evident when integrating ice storage. Both ice storage systems with glazed collectors significantly reduce system energy use, with maximum savings between 17% and 22% in comparison to the air water HP base case. Unglazed collectors show more limited potential because of higher thermal losses during the winter months. Ice storage again has a significant impact on energy use for cooling. By producing ice simultaneously while supplying hot water to the building, cooling energy use is either greatly diminished (82% reduction, Montreal), or completely eliminated (Calgary).

Calgary and Montreal results are quite similar, despite Calgary having greater solar potential. This is related to the collector orientation and solar loop operations. All collector systems are roof integrated, at a slope of 40°. However, as Calgary (N51°) is further north than Montreal (N45°), these collectors receive less intense beam radiation during the winter. Additionally, since solar collector operations are limited in the summer to build up cooling capacity in the ice tank, net summer gains are similar, even though solar availability is higher in Calgary.

Table 8. Annual system energy performance in Montreal and Calgary.

	Montreal					Calgary				
	Base #1	Base #2	SAHP #1	SAHP #2	SAHP #3	Base #1	Base #2	SAHP #1	SAHP #2	SAHP #3
Heat+DHW (kWh)	12770	6141	5948	4700	5520	13463	6179	6056	4906	5434
Cool (kWh)	484	484	181	85	88	283	283	94	0	0
Fans+Pumps (kWh)	990	1486	1546	1573	1566	979	1486	1634	1655	1658
Total Mech. (kWh)	14244	8111	7675	6357	7174	14725	7949	7784	6561	7091

Fig. 2 (a) shows the ice mass in the ice tank for SAHP #2 in Vancouver, while Fig. 2(b) shows the same information for SAHP #3 in Vancouver. Each system uses the ice capacity of the tank from November through February when building thermal demands significantly outweigh available solar energy. While SAHP #2 (with glazed collectors) briefly reaches the maximum allowable ice mass in mid-January, increased thermal gains from the unglazed collectors in SAHP #3 allow this system to use a significant portion of the ice storage while avoiding this upper limit. This is especially important because it allows this system to operate throughout the year without use of the auxiliary heating elements.

Ice buildup during the summer months is part of the defined cooling mode. By limiting solar loop operations, a large quantity of ice and cold fluid can be stored and used as necessary at a cooling coil. This mode also allows for the production of ice during typical DHW operations, eliminating the need for additional heat rejection equipment for all cities except Montreal. While central cooling is minimal in Vancouver, small draws on the ice tank are seen in July and August when natural ventilation is insufficient to keep space temperatures below 24°C.

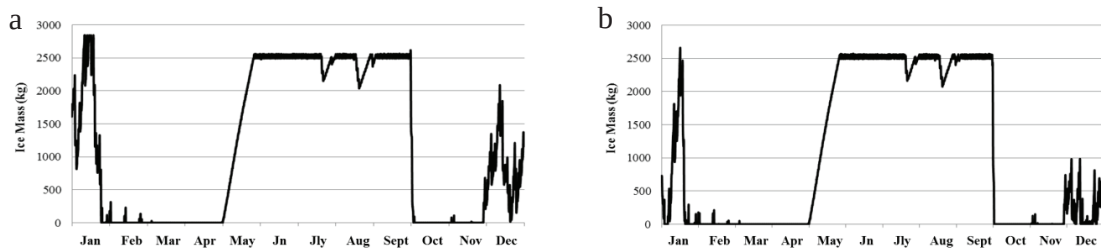


Fig. 2. (a) Ice mass for SAHP #2, Vancouver; (b) Ice mass for SAHP #3, Vancouver.

6. Conclusion

This paper presents an analysis of design options for an innovative solar heat pump concept integrating cool thermal storage. Simulation tools were used to examine a variety of design options, including storage and collector type. Each system was integrated with high performance housing models in Halifax, Montreal, Calgary, and Vancouver. Results show that the proposed solar systems offer strong energy savings potential, with peak reductions for mechanical system energy use ranging from 17% to 28% in comparison to a cold climate air-water heat pump. Unglazed collectors offered promising performance in the milder climates of Halifax and Vancouver, presenting a potential pathway for reducing system capital costs. Ice storage also significantly reduced energy use for cooling, and in many cases eliminated the need for additional heat rejection equipment.

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